

China Solar

Takeaways from SNEC 2026

Equities Energy Equipment & Services

China

- ◆ Space-based solar products still a niche segment, residential ESS drew the most attention; solar supply chain was quiet
- ◆ Solar equipment exports are still restricted as most companies wait rather than seek an SE Asia workaround
- ◆ SST products for AIDC are emerging as the next key battleground

Key takeaways: We attended SNEC in Shanghai on 3–4 June, meeting with investors and engaging with leading solar companies across space solar, residential energy storage systems (ESS), inverters and related segments. Below are our key takeaways:

- ◆ **Space solar:** Still niche at SNEC, but positioned as ‘flagship’ by exhibitors. Solar-cell technology is shifting beyond GaAs towards silicon and tandem. Lightweight covers, such as space colourless polyimide film (SCPI), are a key differentiator. Commercial orders more likely around 2027 after multi-round validation.
- ◆ **Solar equipment:** Near-term outlook is driven more by policy clarity or enforcement than end-demand. Most companies downplayed US export exposure and rejected exporting via Southeast Asia as a workaround. They claimed to adhere strictly to domestic guidance in China.
- ◆ **Residential solar + ESS:** One of the highest-traffic segments. Hardware is commoditising, shifting differentiation to channels and software/service. AI-enabled energy management is becoming baseline, and all-in-one integrated systems are proliferating to simplify installation and user experience.
- ◆ **AIDC & compute/power integration:** ‘Green power for AIDC’ was a major theme, with ecosystem focus on DC coupling and system-level integration. Solid-state transformers (SST) are emerging as a key battleground, but deployment is expected to be gradual, pending proof of lifecycle economics and efficiency.
- ◆ **Solar supply chain:** Noticeably quieter versus storage/AIDC. Pricing remains under pressure and segmented. Cost pass-through is uneven (tax rebate changes partly passed, silver inflation harder), reinforcing ‘optimise/retrofit’ over new builds. Cost-down levers include reduced-silver/copper metallisation.
- ◆ **Energy storage:** Remains the highest-attention growth narrative of some companies. C&I storage is gaining share with improving pipeline stability, while utility-scale is more crowded and price-competitive, pushing some to overseas/niche structures. Margin protection hinges on pricing discipline and the ability to pass through volatile battery input costs.

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Takeaways from SNEC 2026

We participated in SNEC in Shanghai on 3-4 June to visit some of the leading solar companies. Below is a summary of the discussions with company management, experts and investors we met during the tour.

Space solar

- ◆ **Space-based solar product demonstration:** The number of exhibitors explicitly focused on space solar was relatively limited, but those that did show space-related products tended to place them at the centre of the booth narrative (a 'flagship storyline' rather than as a side topic).
- ◆ **Technology direction:** Industry discussion is moving beyond GaAs-only dependence towards silicon-based flexible solutions and tandem concepts (e.g., silicon + perovskite), with >30% efficiency targets cited versus c20% for current silicon space cells.
- ◆ **Lightweight materials as a differentiator:** A recurring theme was replacing traditional glass cover layers (often cited around c110 µm) with ultra-thin polymer/film-based covers (<c25 µm), such as SCPI, to reduce mass while improving flexibility and durability. Moreover, SCPI can bend 360° and withstand c10,000 bends without damage while the ultra-thin glass (UTG) cracks after c20 bends.
- ◆ **Commercialisation timeline:** Industry expectations implied multi-year validation cycles (ground + in-orbit testing). A typical validation cycles of 3–6 months per round with multiple in-orbit trial opportunities (e.g., 20+ launch windows) were referenced to accelerate iteration. Meaningful order conversion is more likely around 2027.
- ◆ **Conservative around SpaceX's 100GW per year of solar capacity goal:** Following our earlier report on space solar ([Space-based Solar](#), 1 June 2026), we found that messaging from the SNEC tour implied conservative attitudes from both companies and investors regarding SpaceX and Tesla's 100GW per year of solar manufacturing capacity goal in three years in the US.

Solar equipment

- ◆ **US export sensitivity:** Many equipment companies downplayed or denied direct exports to the US, with a 'wait-and-see' stance tied to evolving policy interpretation and enforcement.
- ◆ **Southeast Asia not an alternative route:** Companies denied the alternative export route from Southeast Asia to circumvent the domestic restrictions.
- ◆ **Policy uncertainty is the key variable:** Conversations suggested that the near-term swing factor is regulatory clarity and enforcement, not end-demand. Companies emphasised strict adherence to domestic guidance while awaiting official announcements.
- ◆ **Demand shifting by region:** Export opportunities were described as rebalancing away from parts of Southeast Asia (affected by trade measures) towards Middle East / North Africa / Türkiye and other markets.
- ◆ **Retrofit cycle is the anchor:** With cumulative TOPCon capacity cited at 800GW+, the equipment cycle is increasingly upgrade-led rather than greenfield-led. Typical retrofit packages for upgrading old TOPCon to new TOPCon were cited at cRMB30m per GW, lifting module efficiency from 620W to 650W+.
- ◆ **Capex reference points:** New TOPCon line equipment investment was cited at cRMB120-140m per GW. BC line equipment was cited as <RMB200m per GW, with equipment value share in high-power BC lines often described as c40-50% of total line value.

Residential solar + energy storage system (ESS)

- ◆ **Exhibition heat:** Residential storage was one of the busiest segments at the show, with consistently high booth traffic.
- ◆ **Low perceived entry barriers:** The segment looked easy to enter from a product standpoint as hardware differentiation is narrowing, accelerating competition and pushing differentiation towards channels, certification, delivery, and software service.
- ◆ **AI as a standard feature:** Many vendors positioned AI-enabled energy management (tariff arbitrage, smart scheduling, predictive control, VPP participation) as a baseline capability rather than a premium add-on.
- ◆ **All-in-one products proliferating:** Integrated 'all-in-one' systems were widely launched or previewed, signaling a shift towards simplified installation and packaged user experience.

AIDC & compute/power integration

- ◆ **A show highlight:** 'Green power for AIDC' was heavily promoted via renderings, demo boards, and reference architectures, indicating strong ecosystem attention.
- ◆ **Solid-state transformer (SST) momentum:** SST solutions for data-centre power are emerging as a focal point. Multiple players are developing SST-based power solutions for AIDC, and competitive intensity appears real and rising.
- ◆ **Architecture focus:** Discussions frequently centred on DC coupling and system-level integration, aligning with data-centre power distribution trends.
- ◆ **Longer adoption cycle:** Despite strong interest, deployment is expected to be slower, driven by the need to prove lifecycle economics, efficiency, PUE impact, reliability, and standardisation.

Solar supply chain

- ◆ **Relatively quiet:** Compared with storage and AIDC themes, the traditional solar supply-chain narrative felt noticeably cooler at the exhibition.
- ◆ **Pricing remains pressured and segmented:** Mainstream utility-scale pricing was discussed around RMB0.69-0.75/W, while distributed/overseas mixes could clear at RMB0.85-1.00/W. Very low-priced orders were described as more concentrated earlier in the year.
- ◆ **Cost pass-through is imperfect:** Export tax rebate changes were partially passed through in new contracts, while silver cost inflation was harder to fully transfer, keeping margin management tight.
- ◆ **'Optimise and retrofit' over 'build new':** The focus is on process optimisation and cost-down, including reduced-silver/copper-based metallisation. Cost-down impact was cited at RMB0.03-0.05/W under silver prices around RMB18k-19k/kg.
- ◆ **Copper-based metallisation is a key cost-down lever, but ramp takes time:** Industry commentary highlighted copper paste / reduced-silver approaches as a major next step. Initial deployment targets were discussed at the single-digit GW level, implying a gradual scale-up rather than an overnight switch.
- ◆ **Ongoing shakeout:** Industry commentary continued to point to low utilisation rates (often cited at $\leq 50\%$), implying a prolonged capacity-clearing process.

Energy storage

- ◆ **Attention remains elevated:** Storage drew more traffic and mindshare than modules, reinforcing its role as the current "growth and storytelling" engine.

- ◆ **C&I storage gaining share:** Commercial & industrial storage was repeatedly framed as a fast-growing segment, with improving shipment stability and project pipelines.
- ◆ **Utility-scale is competitive:** Large-scale storage was described as more crowded and price-competitive, pushing some participants to prioritise overseas or niche project structures.
- ◆ **Pricing discipline and cost pass-through:** With battery input costs fluctuating, the ability to adjust pricing while protecting margins was highlighted as a key operational differentiator.

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